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Why Steering Works: Toward a Unified View of Language Model Parameter Dynamics

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Abstract

Methods for controlling large language models (LLMs), including local weight fine-tuning, LoRA-based adaptation, and activation-based interventions, are often studied in isolation, obscuring their connections and making comparison difficult. In this work, we present a unified view that frames these interventions as dynamic weight updates induced by a control signal, placing them within a single conceptual framework. Building on this view, we propose a unified preference-utility analysis. This analysis separates control effects into two components: preference, defined as the tendency toward a target concept, and utility, defined as coherent and task-valid generation. Both components are measured on a shared log-odds scale using polarity-paired contrastive examples. Across methods, we observe a consistent trade-off between preference and utility: stronger control increases preference while predictably reducing utility. We further explain this behavior through an activation manifold perspective, in which control shifts representations along target-concept directions to enhance preference, while utility declines primarily when interventions push representations off the model's valid-generation manifold. Finally, we introduce a new steering approach **SPLIT** guided by this analysis that improves preference while better preserving utility[1].

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities and are increasingly deployed in real-world applications (Zhao et al., 2023). Growing demands for safety, controllability, and personalization make reliable control over model behavior a central challenge. To address this, prior work has developed diverse paradigms for controlling LLMs, spanning training-time adaptations, such as local weight fine-tuning and parameter-efficient methods like LoRA (Hu et al., 2022b; Ding et al., 2023; Mao et al., 2025), and inference-time interventions, including activation-level steering via hidden-state manipulation (Rimsky et al., 2024; Han et al., 2024; Bartoszcze et al., 2025; Shu et al., 2025).

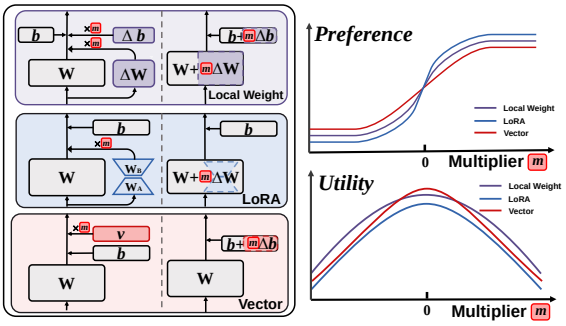

Figure 1: The figure illustrates how different methods operate on the linear layers of the model. We present a unified view in which diverse large language model intervention methods are casted as **dynamic weight updates**. The right panel shows the changes in model utility and preference across different control methods under varying intervention multipliers. Further details are provided in Section 3.

Despite their empirical success, these approaches are often studied in isolation, under different assumptions, objectives, and evaluation protocols. This fragmentation hinders rigorous comparison and obscures shared failure modes. In this work, as shown in Figure 1, we mathematically observe that local weight fine-tuning, LoRA, and activation-level steering can all be formulated as instances of a common *dynamic weight update* framework (Eq. 1). Building on this unified perspective, we introduce a preference–utility analysis and show that, across methods instantiated within this framework, both preference and utility exhibit consistent, predictable patterns as control strength varies.

$$\mathbf{h}_{i+1} = (\mathbf{W} + m_1 \Delta\mathbf{W})\,\mathbf{h}_i + (\mathbf{b} + m_2 \Delta\mathbf{b}). \quad (1)$$

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[1]https://github.com/zjunlp/EasyEdit/blob/main/examples/SPLIT.md.

Note that a particularly challenge in controlled text generation is the **trade-off between enforcing the target concept and preserving task validity**: as control strength increases, the target attribute is amplified, but undesirable side effects—such as incoherence, instruction violations, or context drift—also become more frequent, reducing overall task effectiveness. Moreover, because control quality is typically evaluated via realized outputs, degradation in task validity can confound assessments and obscure the intended concept signal. Guided by this mechanistic understanding, we propose a training objective that explicitly optimizes preference while preserving utility, and experimentally demonstrate that it achieves superior performance.

Our contributions are as follows:

- **Unified View.** We propose a unified view of *dynamic weight updates* that casts local weight fine-tuning, parameter-efficient fine-tuning (e.g., LoRA), and activation interventions (steering) into a common intervention form. Building on this view, we introduce a unified preference–utility analysis and show that, across methods instantiated within the dynamic-update framework, both preference and utility exhibit consistent regularities as control strength varies.

- **Preference–Utility Analysis.** We introduce an *activation manifold* hypothesis and analyze preference and utility under this assumption, suggesting that preference is jointly determined by (i) the projection onto a target-preference direction and (ii) activation validity, which degrades as representations deviate from the manifold, while utility degradation is primarily driven by this off-manifold deviation and the resulting activation invalidation. We further derive two quantitative relationships between the preference log-odds and m, and between the utility log-odds and m, and validate them with high-R^2 fits.

- **New Steering Method.** Guided by this mechanism, we introduce **SPLIT**, a training objective that explicitly optimizes preference while preserving utility, and demonstrate that it achieves better overall performance.

2 Preliminary

2.1 Intermediate Representations in LLMs

During the forward propagation of intermediate layers in LLMs, several key representations occur at specific points in the computation, such as FFN outputs, residual stream states, and linear projections within the attention mechanism (Query, Key, Value, and final output). Ignoring the effects of Layer Normalization[2], these representations can be uniformly expressed as the output of an affine transformation:

$$\mathbf{h}_{i+1} = \mathbf{W}\mathbf{h}_i + \mathbf{b}, \tag{2}$$

where $\mathbf{h}_i$ and $\mathbf{h}_{i+1}$ denote the input and output representations of a linear layer, and $\mathbf{W}$, $\mathbf{b}$ are its weights and biases.

For example, in an FFN block, the up-projection is computed as $\mathbf{h}_{\text{mid}} = \mathbf{W}_{\text{up}}\mathbf{h}_{\text{in}} + \mathbf{b}_{\text{up}}$, followed by a non-linear activation, $\mathbf{h}_{\text{mid,act}} = \sigma(\mathbf{h}_{\text{mid}})$, and then the down-projection is computed as $\mathbf{h}_{\text{out}} = \mathbf{W}_{\text{down}}\,\mathbf{h}_{\text{mid,act}} + \mathbf{b}_{\text{down}}$. Similarly, the Q, K, V, and output projections in the attention module follow the same affine form as in Eq. 2.

2.2 Parameter Update

We consider two parameter adaptation methods for large language models: Low-Rank Adaptation (LoRA) and local weight fine-tuning.

LoRA LoRA freezes the original weight matrix $\mathbf{W}$ and introduces a trainable low-rank update $\Delta\mathbf{W} = \mathbf{B}\mathbf{A}$, where $\mathbf{A} \in \mathbb{R}^{r\times k}$, $\mathbf{B} \in \mathbb{R}^{d\times r}$, and the rank $r \leq \min(d, k)$. At inference, the adapted weights are given by $\mathbf{W} \leftarrow \mathbf{W} + \Delta\mathbf{W}$. In its canonical form, LoRA applies only to the weight matrix while keeping the bias term $\mathbf{b}$ fixed, although extensions exist that also adapt biases.

Local Weight Fine-tuning Local weight fine-tuning updates parameters within a restricted subset of the network, leaving all other parameters frozen. It can be applied to any layer or parameter type, with full-parameter training representing the special case where the subset covers the entire model. A generic update for the weight matrix $\mathbf{W}$ and bias vector $\mathbf{b}$ can be expressed as: $(\mathbf{W}, \mathbf{b}) \leftarrow (\mathbf{W} + \Delta\mathbf{W},\ \mathbf{b} + \Delta\mathbf{b})$. In our experiments, parameter updates are applied only to the **MLP down-projection** layer.

[2]Layer Normalization placement varies across architectures; we omit it here for analytical simplicity.

2.3 Activation Steering

Activation Steering Activation steering modifies intermediate representations during inference by adding a steering vector to selected activations. Its mathematical form can be written as

$$\mathbf{h}_{i+1} = \mathbf{W}\mathbf{h}_i + \mathbf{b} + m\mathbf{v}, \tag{3}$$

where $\mathbf{v}$ is a predetermined direction and m is a scalar coefficient controlling its magnitude. This approach builds on the *linear representation assumption* that abstract concepts correspond approximately to linear subspaces of representation space.

The steering vector $\mathbf{v}$ can be equivalently expressed as a bias adjustment $\Delta\mathbf{b}$, yielding $\mathbf{b} \leftarrow \mathbf{b} + m\Delta\mathbf{b}$. This formulation highlights activation steering as a special case of dynamic parameter update, closely related to methods such as LoRA and local weight fine-tuning.

From a unified perspective, both parameter updates and activation steering operate by injecting a change vector $\Delta\mathbf{h}$ into intermediate representations during forward propagation, differing only in the mechanism by which $\Delta\mathbf{h}$ is generated. More related works can be found in Appendix 6.

3 Unified View of Dynamic Weights in Inference

We present a unified framework for dynamic interventions during inference, as illustrated in Figure 1. Our unified view has three components: (i) a unified *dynamic weights intervention* view that expresses local weight updates, LoRA, and activation steering as dynamic weight updates, (ii) a unified *measurement* view based on preference/utility log-odds, and (iii) a unified *dynamics observation* showing consistent preference–utility response patterns across intervention forms.

3.1 Unified Dynamic Weight Formulation

We propose a unified framework that encompasses both parameter update methods and activation steering methods, by viewing them as dynamic weight updates. Under this formulation, both can be expressed through a shared affine transformation view of intermediate representations; detailed derivations and formulations are provided in Section 2.

Formally, the dynamic modification of the weight matrix $\mathbf{W}$ and bias vector $\mathbf{b}$ during inference can be written as:

$$\mathbf{h}_{i+1} = (\mathbf{W} + m_1\,\Delta\mathbf{W})\,\mathbf{h}_i + (\mathbf{b} + m_2\,\Delta\mathbf{b}), \tag{4}$$

where $\Delta\mathbf{W}$ and $\Delta\mathbf{b}$ are update terms, and m_1, m_2 are scalar scaling coefficients controlling their magnitudes (Fierro and Roger, 2025). In other words, the original parameters are updated to $\mathbf{W}' = \mathbf{W} + m_1\,\Delta\mathbf{W}$ and $\mathbf{b}' = \mathbf{b} + m_2\,\Delta\mathbf{b}$ before computing the next-layer activation.

When a model weight is modified, the effect can be equivalently interpreted from the activation perspective, as a change to the activation at the corresponding position. In this view, diverse intervention methods are unified as adding a change term to the activation:

$$\Delta\mathbf{h} = m_1\,\Delta\mathbf{W}\,\mathbf{h}_i + m_2\,\Delta\mathbf{b}. \tag{5}$$

Under this unified view, local weight fine-tuning, LoRA, and activation steering are all specific instances, differing only in which components are updated: local weight fine-tuning modifies both $\mathbf{W}$ and $\mathbf{b}$; LoRA modifies $\mathbf{W}$ via low-rank factors; activation steering modifies only $\mathbf{b}$. Table 1 summarizes their affine forms, corresponding activation update, and parameter sizes.

Notably, introducing explicit scaling coefficients extends traditional formulations and enables continuous control over perturbation strength, a capability that plays a central role in our subsequent analysis.

3.2 Unified Analysis View: Preference and Utility Log-Odds

We analyze intervention effects along two complementary dimensions. **Preference** denotes the model's internal inclination toward a target concept, independent of whether the model completion is well-formed. For the prompt "*Write a short review for this restaurant*", generating "*The food was excellent and the service was wonderful*" indicates a positive preference, while "*The food was terrible and the service was disappointing*" indicates a negative preference. **Utility** denotes the model's task competence that is independent of the target concept. It captures whether the model can produce a task-valid completion that is coherent, relevant to the prompt, and consistent with the requested format. For the same prompt, utility is high when the output is a readable restaurant review, regardless of polarity. Utility is low when the output is incoherent such as "*food food wonderful ??? service 19% ##*", off-topic such as "*Here is a Python script to scrape restaurant data...*", or instruction-violating even if polarity-bearing words appear.

In controlled generation, performance is typically evaluated from the realized outputs. When